Elena de Sanctis

23-08-2003 | Milan, Italy
+39 388 074 2253 | ellidesanctis@gmail.com | www.linkedin.com/in/elena-de-sanctis

Education

Exchange Semester
Linköping University, Linköping, Sweden *Aug. 2026 – Present*
- Exchange semester as part of the MSc in Mathematical Engineering at Politecnico di Milano.
- Relevant coursework: Applied Computational Fluid Dynamics, Fluid Mechanics, Mathematical Models in Biology, Text Mining, Medical Image Analysis

MSc in Mathematical Engineering – Computational Science and Computational Learning
Politecnico di Milano, Milan, Italy *Sep. 2025 – Present*
- Relevant coursework: Real and Functional Analysis, Advanced Partial Differential Equations, Numerical Analysis for Partial Differential Equations, Algorithms and Parallel Computing, Advanced Programming for Scientific Computing, Fluid Dynamics, Computational Fluid Dynamics
- Member of AIM – Association of Mathematical Engineers

BSc in Mathematical Engineering
Politecnico di Milano, Milan, Italy *Sep. 2022 – Sep. 2025*
- Relevant coursework: Probability Theory, Statistical Inference, Numerical Analysis, Linear Algebra, Partial Differential Equations, Functional and Complex Analysis
- Member of AIM – Association of Mathematical Engineers

Projects

Phase-Field Fracture Propagation | *C++, Material Point Method, Explicit Solvers* Spring 2026 – ongoing
- Built a C++ solver for brittle fracture coupling the equilibrium equation with a phase field damage model on a Material Point Method discretisation, developed incrementally from the 1D quasi-static case to the 2D time-dependent one, with a spectral decomposition of the elastic energy so that only tensile strain drives damage.
- Reformulated the implicit schemes of the reference papers into a fully explicit, matrix-free algorithm: explicit momentum update under a CFL time step, and a damped Jacobi iteration for the elliptic phase field equation, removing any global linear solve. Currently extending it to a parallel implementation.
- Reproduced the quasi-static single-edge-notched shear test and the dynamic crack branching benchmark from the reference literature, and verified the mechanical solver independently against the analytical shear stiffness, matching it to five significant digits.

External Aerodynamics of the DrivAer Car | *ANSA, Ansys Fluent, ParaView, Python* Sep. 2026 – Oct. 2026
- Prepared the DrivAer geometry in ANSA: CAD cleaning, surface meshing and a half-model domain with moving ground, wheel contact patches and three nested refinement regions (25, 75 and 200 mm).
- Wrote a parametric Scheme script for Ansys Fluent Meshing that automates the volume mesh: size fields, 10 prism layers with different methods on vehicle and ground, and a 9.3-million-cell poly-hexcore mesh.
- Automated the steady RANS simulation (realizable k–ε, 20 m/s, rotating wheels) in Fluent and the post-processing in ParaView (Python), run in batch mode on the NSC Sigma HPC cluster.

Smoke Dispersion from a Vehicle Fire | *Ansys Fluent, MATLAB, ParaView, HPC* Sep. 2026 – Oct. 2026
- Simulated the smoke from a bus fire on the Linköping University campus to identify the buildings most at risk, running steady RANS simulations (realizable k–ε, 20 million cells) on the NSC Sigma cluster with scripted Fluent journals.
- Reduced 20 years of hourly wind data to five representative cases with k-means clustering in MATLAB, and combined them into probability-weighted exposure maps.
- Compared a reacting combustion model with a passive-scalar model, showing that the simpler model captures the smoke direction but not its rise or chemistry. Co-authored a scientific paper and presented a poster.

Technical Skills

Programming: C/C++, MATLAB, R, LaTeX
Software: Ansys Fluent, ANSA, ParaView, HPC clusters (Slurm)

Additional Information

Languages: Italian (Native), English (B2, Cambridge FCE; MSc taught entirely in English)
Interests: Sport, Fashion
Volunteering: Leo Club Rieti Host (youth branch of Lions Clubs International) – Vice-President 2021/22, President 2023/24, member since 2020